

Explaining Debt Recovery Using an Endogenous Bankruptcy Model

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There are two main elements for describing credit risk that are relevant to the valuation and hedging of risky debt and credit derivatives: the probability of default and the outcome of default (i.e., recovery (or loss) upon default). Earlier studies on credit risk have focused their attention on either the characterization of the timing of default to infer the likelihood of default or the pricing of defaultable debt while modeling the joint product of default probability and recovery. In many of these studies, recovery is assumed to be static. However, empirical evidence suggests that there exists a large cross-sectional variation in debt recovery that is unexplained.

Recently, a growing body of literature has been studying the determinants of debt recovery. One adopted approach is to identify factors that influence bankruptcy events, such as debt characteristics, firm's financial condition, industry distress, and the economic environment, and then use statistical methods to estimate the recovery. Many studies have adopted this approach.¹ An alternative approach is to estimate model-implied recovery from default-related instruments, such as credit default swaps and corporate debt.² This approach provides the risk-neutral measure of debt recovery relevant to pricing credit instruments.

Merton [1974] identifies that a corporate security can be viewed as an option on firm

assets and hence can be priced through option pricing theory. Building on this basic logic, many variations of the structural models have been developed. These models can be attributed to three basic types: exogenous default boundary models (such as Black and Cox [1976], and Longstaff and Schwartz [1995]), endogenous default boundary models (such as Geske [1977], Leland [1994], and Leland and Toft [1996]), and strategic debt service models (such as Mella-Barral and Perraudin [1997] and Fan and Sundaresan [2000]).

Prior studies have empirically estimated the model parameters to explain the default likelihood and/or to price corporate bonds (for example, Eom et al. [2004], Ericsson and Reneby [2005], Suo and Wang [2006], and Huang and Huang [2012]). However, the findings from these studies are mixed on model performance. Several studies point out that credit spreads generated by structural models are, in fact, too low to explain the observed yield spreads on corporate bonds, while others find they are relatively successful in predicting default probabilities and calculating hedge ratios (Schaefer and Strebulaev [2008]).

None of the existing studies make an attempt to use structural models to explain debt recovery. The exogenous default boundary models assume that the default barrier is exogenously chosen and creditors recover a fixed fraction of the debt face value in default. Therefore, according to these

models, debt recovery is predetermined. In contrast, the endogenous default boundary models suggest that managers choose the timing of bankruptcy, whereby the bankruptcy threshold is endogenously determined to maximize the ex ante value of equity. Shareholders continuously service their debt as long as the firm asset value is above the endogenous threshold level. When the asset value falls below the threshold, bankruptcy is filed, a deadweight loss associated with bankruptcy costs is deducted from the firm asset, and the remaining assets are distributed to the debtholders. Prior studies show that these models produce smaller errors than other types of structural models in pricing defaultable bonds.³ The intuition behind these models is consistent with the fact that share-value-maximizing managers predominantly make bankruptcy-filing decisions.⁴

The endogenous default boundary allows us to estimate what creditors expect to recover when a firm chooses to declare bankruptcy before debt matures. In this article, we intend to explain debt recovery observed in the market by using the endogenous bankruptcy model developed in Leland and Toft (LT) [1996] for two reasons. First, their model considers the finite maturity of debt and provides a closed-form solution for pricing equity, which in turn allows us to estimate the parameters of the unobservable firm asset from equity prices. Second, prior studies have demonstrated that their model performs better than other models at determining the default boundary as well as predicting default likelihood (for example, Leland [2004], Suo and Wang [2006], and Davydenko [2012]).⁵

In this study, we apply the maximum likelihood estimation (MLE) method to estimate the parameters for the process that governs the evolution of asset value in the LT model from equity prices and the average costs of bankruptcy that are suggested in the literature. Given this, we are able to estimate the endogenous bankruptcy barrier, which represents the value that is distributed to debtholders after bankruptcy costs are deducted, for a sample of 276 Chapter 11 filings by large U.S. firms from 1996 to 2007. This sample was compiled from a variety of data sources that required intensive manual data collection.

We find that the recovery estimated from the model using four quarters of accounting information and equity prices prior to bankruptcy filing has strong explanatory power on firm-level recovery, calculated as the weighted-average recovery of defaulted bank loans and bonds. Our results hold after we account for firm

characteristics, industry distress, and macroeconomic conditions, as well as the application of alternative expected bankruptcy costs. Our findings suggest that the estimated recovery from the LT model is a useful tool for explaining the observed debt recovery.

The strong predictive power of the endogenous default boundary from the LT model on debt recovery could be due to, but not limited to, two explanations. First, endogenous bankruptcy models emphasize the manager's role in choosing the default barrier by maximizing shareholders' wealth, which is consistent with empirical evidence on bankruptcy-filing decisions. Second, the stock price may contain information related to debt recovery, and the success of the LT model is simply a reflection of the explanatory power of equity prices. To eliminate the second explanation, we first examine whether pure stock returns or prices prior to bankruptcy can explain debt recovery but find no supporting evidence.

Next, we investigate whether agency problems and heterogeneous bankruptcy costs, two factors that are not considered in the endogenous bankruptcy models but are widely documented in the literature, can weaken the explanatory power of the model. In consideration of agency problems, we separate our sample firms by using CEO stock holdings and pay-for-performance sensitivities in the year prior to bankruptcy filing and perform subsample analysis. As agency conflicts are not considered in the LT model, the timing of bankruptcy may not be chosen optimally by managers. In firms with more severe agency problems, default barriers are less likely to be determined through the maximization of shareholders' equity, and thus the explanatory power of the endogenous default barrier on market recovery can be weakened.

Our results confirm this hypothesis. The explanatory power of this model greatly improves for firms with less severe agency conflict, as evidenced by larger CEO equity ownership and higher pay-for-performance sensitivities tied to the CEO's wealth. This provides the first supporting evidence that the explanatory power of the model comes from the endogenous default barrier optimally chosen by value maximizing managers.

Next, we examine whether the large difference between the estimated and realized bankruptcy costs can weaken the power of our test. We separate our sample firms into two groups based on the length of time a firm stays in bankruptcy, which is used as a proxy for bank-

ruptcy costs (in the absence of this specific information). The bankruptcy duration of the first group of firms lies within the interquartile range of the duration spectrum. The second group of firms has bankruptcy duration lying either below the 25th percentile or above the 75th percentile of the duration distribution. Our results show that the explanatory power of our model is much stronger for the first group of firms, suggesting that the estimated recovery explains market recovery better when the realization of bankruptcy costs is closer to the ex ante cost parameter assigned to the estimation. Additionally, we endogenize bankruptcy costs by using duration and firm characteristics and find our results remain unchanged.

This article makes contributions to the current literature on debt recovery in several areas. First, we show that the recovery rates derived from an endogenous bankruptcy model can be used to predict recovery if a firm files for bankruptcy in the foreseeable future. In contrast to most previous statistical studies on debt recovery, our results suggest that managers play an important role in endogenously determining the timing of bankruptcy, which affects the default boundary and therefore debt recovery. We find that this estimated recovery contains more information for the market recovery than other common sets of economic and firm-specific factors.

Furthermore, we identify that agency problems may distort the optimal default barrier chosen by shareholders. Given this, our study suggests that when building credit risk models, one should incorporate these more realistic assumptions—such as agency conflicts—in determining default boundaries. This study also provides a useful tool for portfolio managers and risk managers to predict firm-level recovery in valuing and hedging risky debt and credit derivatives.

The rest of the article is organized as follows. First, we provide a brief review of the Leland–Toft model and the estimation procedure. We then describe the data sample and provide the main results and discussions. The final section presents our conclusions.

THE LELAND–TOFT MODEL AND ESTIMATION

Model

Leland and Toft [1996] developed a debt pricing model that relates equity and bond values to the firm asset risk, bankruptcy cost, tax benefit of debt, and risk-

less interest rate. A firm defaults whenever its asset value falls below a predetermined bankruptcy threshold. Different from exogenous default boundary models, the bankruptcy threshold in their model is endogenously determined by managers on behalf of equityholders. The firm asset value is unobservable and is assumed to follow a geometric Brownian motion. When bankruptcy occurs, a proportion w of firm asset is lost due to bankruptcy costs, with the remaining assets being distributed to creditors according to the priority rule. Leland and Toft show that for a chosen default trigger level V^* , the debt and equity value can be expressed analytically. A brief mathematical description of the LT model and the expression for V^* are provided in Appendix A.

The LT model assumes that managers of a firm act in the best interests of shareholders. The default barrier is chosen by the management to maximize the equity value and can be expressed in terms of asset volatility, firm leverage, coupon rate, debt maturity, corporate tax rate, bankruptcy cost, and the risk-free interest rate. The trigger V^* is the asset level such that the change in the value of equity equals the additional cash that must be provided by equityholders to keep the firm solvent. When $V > V^*$, equity has a positive value and the expected equity value appreciation exceeds the cash flow (if any) that must be contributed for the firm to fulfill its debt obligations. One should never observe $V < V^*$ for a solvent firm because equity value appreciation would be less than the contribution required from equityholders to keep the firm solvent. Unlike the Merton model and some of its extensions, the default barrier in the LT model can be higher or lower than the principal amount of its outstanding debt. This is because the tax deductibility of interest payments may be lost when bankruptcy is declared.

When a firm files for bankruptcy, a fraction, w , of the asset value is lost due to direct and indirect bankruptcy costs. The recovery rate is the remaining value $(1 - w)V^*$ that debtholders receive as a fraction of the debt face value F , that is,

$$\text{LT Recovery} = \frac{(1 - w)V^*}{F} \quad (1)$$

This recovery is a reasonable estimate of the firm-level recovery, illustrated as follows. The default boundary, V^* , measures the value of the asset at bankruptcy before the deadweight costs are deducted. There

are two components in asset value at bankruptcy: the collateral used for secured debt and the remaining assets available for distribution to unsecured lenders. Although bank lenders may not be able to seize the collateral upon bankruptcy filing due to automatic stay, they do have the ultimate right to claim the collateral. The value of secured bank loans should thus be tied closely to the value of collateral.⁶ Unsecured lenders, on the other hand, share the remaining assets after collateral is liquidated and paid to secured bank lenders. Therefore, the recovery rate of bank loans and bonds with bankruptcy costs w^* borne by unsecured debt holders can be expressed as

$$\begin{aligned}\text{Recovery}_{\text{Bank}} &= \text{Collateral} / \text{Face}_{\text{Bank}} \\ \text{Recovery}_{\text{Bond}} &= \text{Remaining Assets} \times (1 - w^*) / \text{Face}_{\text{Bond}}\end{aligned}$$

respectively. Firm-level recovery can be calculated as the weighted-average recovery of both loans and bonds, expressed as

$$\begin{aligned}\text{Recovery}_{\text{Firm}} &= \frac{\text{Face}_{\text{Bank}}}{\text{Face}_{\text{Bank}} + \text{Face}_{\text{Bond}}} \frac{\text{Collateral}}{\text{Face}_{\text{Bank}}} \\ &+ \frac{\text{Face}_{\text{Bond}}}{\text{Face}_{\text{Bank}} + \text{Face}_{\text{Bond}}} \frac{\text{Remaining Asset} \times (1 - w^*)}{\text{Face}_{\text{Bond}}}\end{aligned}$$

that is,

$$\begin{aligned}\text{Recovery}_{\text{Firm}} &= \frac{\text{Collateral} + \text{Remaining Assets}(1 - w^*)}{\text{Face}_{\text{Bank}} + \text{Face}_{\text{Bond}}} = \frac{(1 - w)V^*}{F}\end{aligned}\quad (2)$$

The illustration above shows that the recovery rate calculated from the endogenous default barrier model provides a reasonable estimate of firm-level recovery rate.

Estimation Method and Model Inputs

In order to calculate the default trigger level, we need to know the asset growth rate and volatility as well as firm leverage. However, since the firm asset is not tradable and thus its historical value is unobservable, we need to estimate the model parameters from observ-

able security prices. In this study, we adopt the MLE method, which allows us to estimate the parameters of the underlying asset value process simultaneously from observable equity prices. The MLE method has been shown to outperform other methods in terms of both the lack of bias and efficiency (see Ericsson and Reneby [2005] and Pan and Singleton [2008]).⁷

The main goal of the MLE approach is to maximize the log-likelihood function of firm security prices to obtain the model parameters. In structural models, firm asset value is assumed to follow a geometric Brownian motion, and hence the logarithm of asset value follows a Gaussian distribution. The unknown density function of equity prices can be computed as the product of a Jacobian function and the density function of asset value (Duan [1994]). Given this, the likelihood function related to stock prices can then be easily obtained (see Appendix B for details). By maximizing the likelihood function, we can obtain the model parameters μ_v and σ_v , which represent the growth rate and volatility of the asset value, respectively. With the estimated parameters, we can in turn compute the implied default barrier V^* and LT recovery expressed in Equation (1).

We carefully match the quarterly balance sheet information to the corresponding daily equity data for each firm during the 252 days preceding Chapter 11 filing. Since the contemporaneous accounting information is difficult for creditors to obtain, we follow prior literature by using firm-level accounting information in the fiscal quarter immediately prior to the equity observation. We employ the book value of total debt to proxy for the debt face value, F , in the model. The coupon payout, c , is the dollar-weighted-average coupon rate of all outstanding debt at the bankruptcy. The initial maturity, T , is estimated as two times the weighted-average maturity of all debt instruments, while the maturity of all bank loans is assumed to be one year.⁸ The corporate tax rate, τ , is assumed to be 35% for all firms. The dividend payout rate is zero for most sample firms.

We select 15% as the deadweight loss in bankruptcy for all firms in our sample, given that Altman [1984] and Andrade and Kaplan [1998] suggest that the cost of financial distress is likely to be in the range of 10% to 20% of asset value.⁹ Finally, the risk-free interest rate, r , is measured by the one-year constant maturity Treasury rate reported by the Federal Reserve Bank.¹⁰ With these inputs, the MLE method can be performed for each firm in our sample, and the asset volatility σ_v is obtained. The

market value of assets can be computed for each daily observation of equity value. With these values in place, the LT recovery can be now calculated.

DATA DESCRIPTION

Data Sample

We start with UCLA–Lynn M. LoPucki Bankruptcy Research Database to draw on the sample of all large U.S. firms that filed for Chapter 11 during 1996–2007. For inclusion in this database, a firm must have had assets worth at least \$100 million at the time of a Chapter 11 filing, measured in 1980 dollars, and must have filed with the U.S. SEC in at least one of the three years prior to filing for bankruptcy. Bankruptcies filed by financial institutions are dropped.¹¹ Our starting sample includes 456 filings. As of January 1, 2011, 439 cases were reorganized, liquidated, or acquired, and 17 cases were dismissed or still pending.

Defaulted debt instruments and recovery.

We collect detailed information on defaulted debt in our bankruptcy sample, including instrument type, seniority, principal, coupon rate, and maturity date from New Generation Research's bankruptcydata.com. We cross-check the debt characteristics with reorganization and liquidation plans confirmed by the U.S. Bankruptcy Courts and Standard and Poor's (S&P) LossStats database.¹² We are able to obtain an initial sample of 607 bank loans and 1,068 bonds for 392 bankrupt entities.

Bankruptcydata.com and Bankruptcy DataSource provide post-petition trading prices of defaulted bonds for a majority of the cases in our data sample.¹³ We use the trading prices at the first month end after Chapter 11 filing to measure the recovery of par value.¹⁴ For bonds with trading prices missing, we turn to the LossStats database to retrieve the trading prices conditional on the actual default date reported by S&P being within one month of Chapter 11 filing. The post-petition bond trading prices are verified with Datastream whenever possible. There are 594 bonds with post-petition trading price available from all of these sources.

S&P carries post-petition trading prices for a small sample of secured bank loans, as these loans are not frequently traded. Prices are available for only those loans that are traded right after bankruptcy filing. In order to minimize the bias from using bank loan post-petition trading price as a measure of recovery, we turn to the

loan recovery from reorganization or liquidation plans confirmed by courts, retrieved from Bankruptcydata.com and company 8K filings. This measure is often referred to as the "ultimate recovery." Since bank loans are usually short term and accrue interest post-petition, we do not apply discounting to the recovery at Chapter 11 filing.

In order to estimate the firm-level recovery for each case, we need the debt principal amount and trading prices of all debt instruments at bankruptcy. That being said, trading prices are not available for more than 40% of our sample bonds. Given the fact that bonds of the same seniority are usually grouped into one claim class regardless of coupon and maturity at bankruptcy resolution, bonds of the same seniority of a given firm have very similar recovery rates.¹⁵ We use the average recovery of all other bonds of the same seniority of a firm as a proxy if the trading price of a bond is missing.

Furthermore, in order to estimate firm-level recovery for many of our sample firms without complete information on debt recovery, we adopt two (valid) assumptions. First, if there are no seniority peers for a bond within the same firm, we use the average trading price of bonds that are issued by all firms in the same Fama–French 12-industry classification and have the same seniority in the sample. In a similar vein, if a secured bank loan does not have the ultimate recovery, we use the mean recovery of all loans with ultimate recovery available in the sample. Second, senior unsecured and subordinated bank loans tend to have recovery rates similar to those of the public bonds of the same seniority (Cantor et al. [2009]). We group the unsecured bank loans into the corresponding class of unsecured bonds and employ the bond trading price as a proxy for these two loan categories. For a firm to be included in our sample, it must have at least one bond or bank loan with a recovery rate available from various data sources before any proxies are applied.

Firm accounting and security pricing information. Firm accounting information is retrieved from quarterly filings in Compustat or from 10-Q filings through EDGAR when the information is missing from Compustat. Our firm-level control variables include firm size, leverage, profitability, asset tangibility, and number of corporate debt outstanding. Firm size is defined as the logarithm of book assets, deflated to 2008 dollars. Leverage is measured as the ratio of the sum of debt in current liabilities and long-term debt to book

value of assets. Profitability is measured by return on assets (ROA) as calculated by the ratio of earnings before interest, taxes, depreciation, and amortization (EBITDA) to assets. Asset tangibility is defined as property, plant, and equipment (PP&E) to book assets. Number of debt issues is the total number of corporate debt outstanding at the time of bankruptcy. All accounting control variables are taken from the fiscal quarter immediately before Chapter 11 filing.

We rely primarily on CRSP to retrieve daily stock information for the 252 trading days prior to Chapter 11 filing. We turn to Bloomberg and Datastream for additional information whenever CRSP stops coverage of a bankruptcy firm due to delisting. We require a firm to have at least 90 daily trading prices and at least two fiscal quarters of accounting information.¹⁶ After merging equity and accounting information with firm-level recovery based on the relaxed assumptions, we have a final sample of 276 firms with 1,354 debt instruments for our study.

Industry and economic conditions. Following Acharya et al. [2007], we define a dummy variable, *industry distress*, based on the two-digit SIC code to capture the industry condition around bankruptcy filing. Industry distress dummy equals one if the median stock

return for the industry of the defaulting firm in the year before bankruptcy is less than or equal to -30% .¹⁷ Furthermore, we control for industry fixed effects in the regression analysis. S&P500 index returns are used to capture macroeconomic conditions. The impact of S&P500 index returns on debt recovery is expected to be positive since poor macroeconomic conditions reduce buyers' potential ability to pay high prices for defaulted firms.

Sample Overview

Chapter 11 filings are highly cyclical, with 40% of our sample firms clustered during the 2001–2002 period, the recession years after the tech bubble burst. Exhibit 1 plots the mean firm-level market recovery by year of Chapter 11 filing. As expected, the recession years are associated with low debt recovery, while years with fast economic growth are associated with high debt recovery. Exhibit 2 shows average recovery by the Fama–French 12 industry classification (not including the banking and finance industry). Utilities and chemicals are among the industries with the highest overall recovery, while telecommunications and healthcare recover the least for creditors. Our findings are consis-

EXHIBIT 1

Mean Market Recovery by Year of Chapter 11 Filing

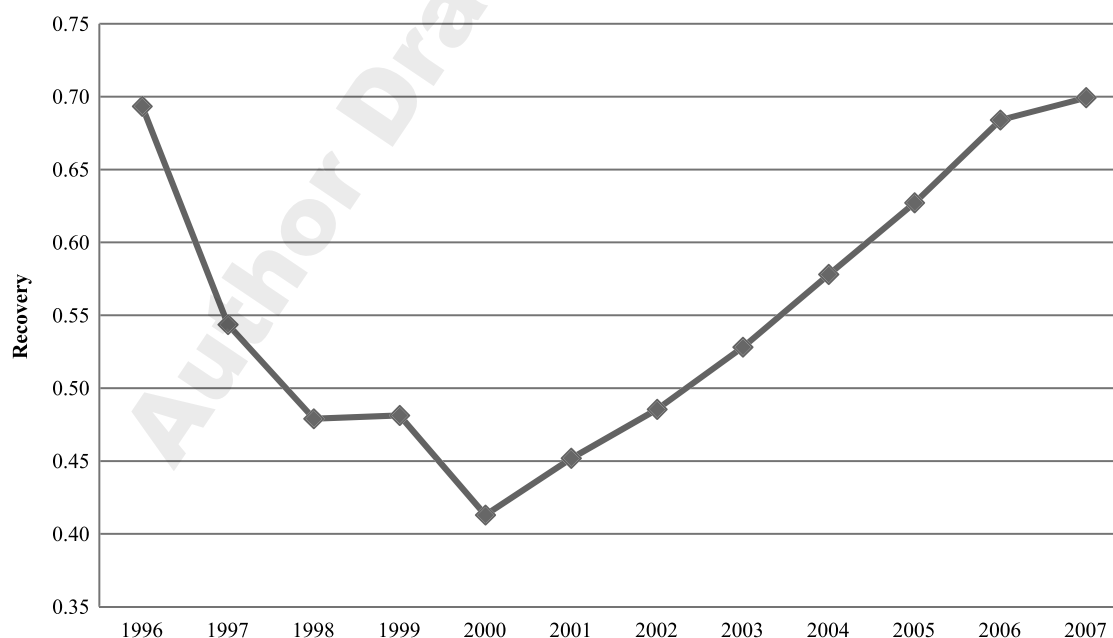
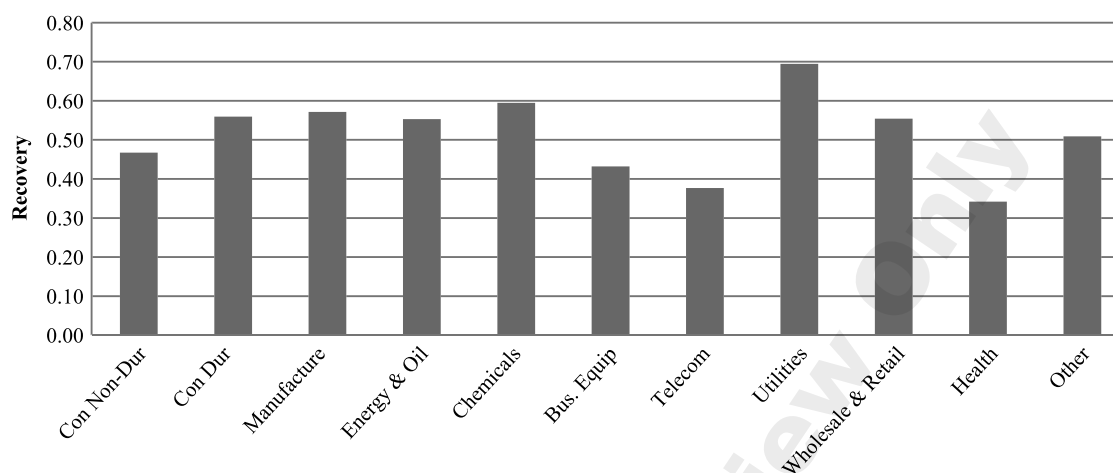


EXHIBIT 2

Mean Market Recovery by Fama–French 12 Industries



tent with those presented in earlier studies (see Altman and Kishore [1996], for example).

We provide summary statistics on bond, firm, industry, and macroeconomic characteristics in Exhibit 3. The mean weighted-average debt maturity is approximately 4.4 years. The coupon rate is 8.1% on average. Meanwhile, a firm generally has four to five debt instruments outstanding at the time of Chapter 11 filing. The median values of the accounting characteristics for our sample firms are: 848 million dollars of assets, 185 million dollars of sales, 0.4% ROA, 68.9% book leverage ratio, and 40.5% tangibility. Compared with the median values of the entire Compustat database, our sample firms are larger and have higher leverage, lower ROA, and higher proportion of tangible assets. Approximately 30% of the Chapter 11 filings are pre-packaged, and another 7% of the filings are caused by fraud. More than 60% of the sample firms emerge as independent companies, and 23% of the firms are liquidated. The average time for firms to stay in bankruptcy is approximately one and a half years, which is close to the mean of all cases in the Bankruptcy Research Database from 1980 to 2009. The industry distress dummy is one for 18% of the cases.

We report the market recovery and the LT recovery as well as μ_v and σ_v from the MLE estimation in the bottom panel of Exhibit 3. The mean (median) of the estimated recovery based on an ex ante bankruptcy cost of 15% for the whole sample is 0.48 (0.53), very close to that of market recovery at 0.49 (0.51). We fail to reject

the null hypothesis that their mean (or median) is not different based on both the t -test and the Wilcoxon signrank test. The mean values of μ_v and σ_v are -0.32 and 0.42 , respectively, indicating our sample firms experienced a large decline of their asset values before bankruptcy filing.

Exhibit 4 presents the implied asset value as well as estimated default boundary of Seitel Inc. before it filed for Chapter 11 on June 6, 2003. This figure shows how the default barrier and implied asset value change through time from 252 to 5 days prior to bankruptcy filing. The time series of the default barrier and the implied asset value are obtained from the structural estimation described in earlier sections. We observe a large decline of firm asset value prior to bankruptcy filing while the default boundary is rather stable. Lastly, firm asset values move very close to the bankruptcy barrier before the filing.

MULTIVARIATE RESULTS

Main Results

Exhibit 5 reports the main multivariate regression results. The variable of interest is *LT recovery*. Besides the firm and industry variables that are commonly used in earlier studies, we control for whether or not a bankruptcy filing is caused by accounting fraud. Fraud cases are associated with greater information asymmetry. The information released after the revelation of a fraud can be

EXHIBIT 3

Summary Statistics of Chapter 11 Firms

This exhibit presents the summary statistics for 276 sample firms. Debt characteristics include: *Maturity*, the weighted average of maturities of all debt instruments with maturity of bank debt assumed to be one year; *Coupon*, the weighted-average coupon rate of all outstanding debt at the time of bankruptcy; *Number of issues*, measured as the total number of debt issues at Chapter 11 filing; *Bank loan*, the proportion of bank loan amount of firm total debt. Firm accounting information is based on the last fiscal quarter before bankruptcy. *Asset* is defined as the book value of assets (in 2008 dollars); *Sales* is the net sales (in 2008 dollars); *ROA* is the ratio of operating income before depreciation to asset; *Leverage* is the ratio of total debt (long-term debt plus debt in current liabilities) to assets; *Tangibility* is the ratio of PP&E to assets. Bankruptcy characteristics include: *Prepack*, an indicator variable equal to one if Chapter 11 is a prepackaged case; *Fraud*, an indicator variable equal to one if Chapter 11 is caused by fraud claims against the company; *Reorganized*, an indicator variable equal to one if a company emerges as an independent firm; *Liquidation*, an indicator variable equal to one if a case is converted to Chapter 7 or liquidated in Chapter 11; and *Years in bankruptcy*, the number of years from petition to confirmation or other disposition of the case (January 1, 2011, for pending cases). Industry variables are: *Industry distress*, an indicator variable equal to one if the median stock return of all other firms with the same two-digit SIC code is less than -30% in the year prior to bankruptcy; *SP500* is the annual return on the S&P 500 Index. *Market recovery* is the firm-level weighted-average recovery of bonds based on distressed trading price and loans based on ultimate recovery. *LT recovery* is the estimated recovery from the Leland-Toft model.

Variable	Mean	Std	P25th	Median	P75th
Firm Characteristics					
Maturity	4.396	3.069	2.651	3.776	5.569
Coupon	0.081	0.028	0.064	0.082	0.099
Number of issues	4.891	4.551	2	4	6
Issue size (in million \$)	1414	3193	258	563	1275
Bank loan	0.351	0.282	0.091	0.340	0.542
Firm Characteristics					
Asset (in million \$)	3234	10026	441	848	2287
Sales (in million \$)	730	3622	94	185	447
ROA	-0.006	0.048	-0.015	0.004	0.018
Leverage	0.768	0.381	0.514	0.689	0.914
Tangibility	0.416	0.235	0.226	0.405	0.600
Bankruptcy Characteristics					
Prepack	0.293	0.456	0	0	1
Fraud	0.069	0.254	0	0	0
Reorganized	0.649	0.478	0	1	1
Liquidation	0.232	0.423	0	0	0
Years in bankruptcy	1.588	1.676	0.522	1.114	1.979
Industry and Economic Conditions					
Industry distress	0.178	0.383	0	0	0
SP500	0.004	0.188	-0.130	-0.101	0.195
Firm-Level Recovery					
Market recovery	0.494	0.221	0.317	0.507	0.662
LT recovery	0.483	0.140	0.411	0.531	0.562
μ_v	-0.319	0.503	-0.508	-0.227	-0.075
σ_v	0.423	0.295	0.222	0.332	0.540

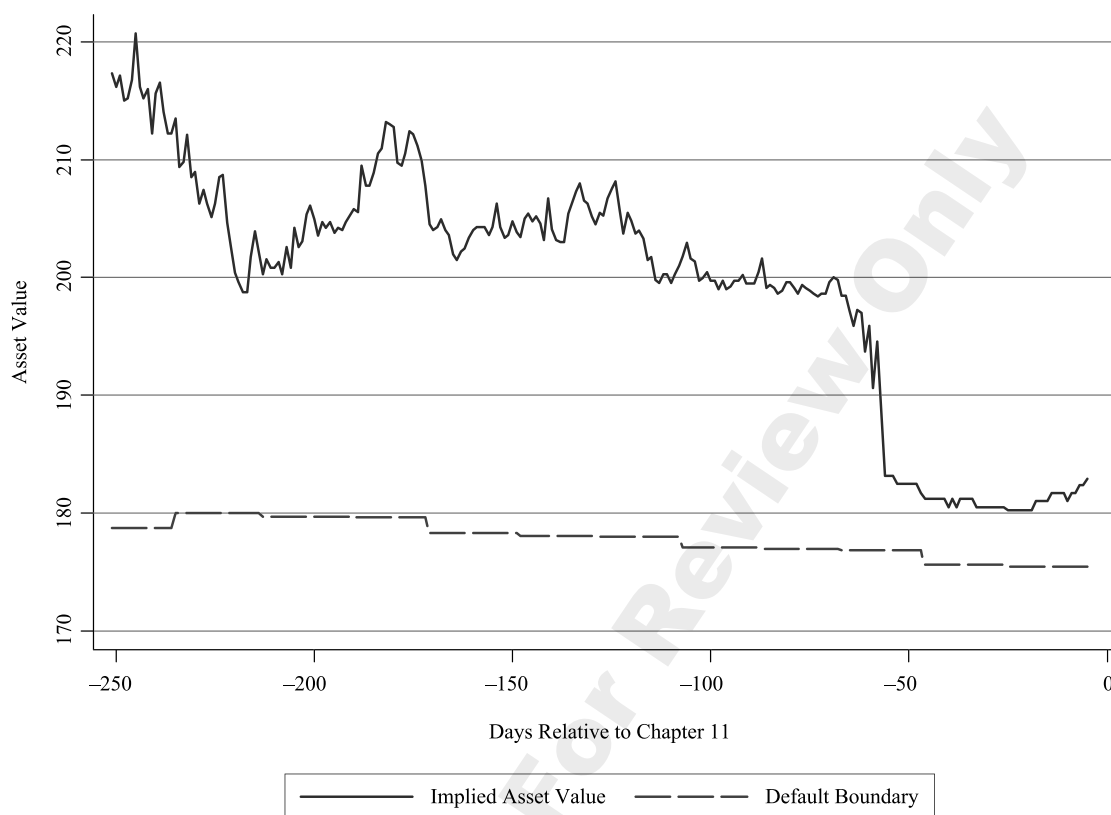
substantial and critical. Therefore, the changes in asset and debt values during bankruptcy filing can be drastic. Moreover, bankruptcy costs are expected to be severe in fraud cases. For instance, firms are likely to suffer a great loss of reputation and growth opportunities after a fraud revelation. As a result, market recovery is expected to be lower in such cases.

We find that the coefficient of LT recovery is positive and significant at the 1% level in all regressions. The

adjusted R-squared with LT recovery alone in Column (2) of Exhibit 5 is higher than the adjusted R-squared with all the firm-level as well as industry and macroeconomic controls in Column (1). This suggests that LT recovery alone has a stronger explanatory power on the variation of firm-level market recovery than do all other common variables documented in previous studies. Column (3) shows that firm-level accounting information, industry distress, and macroeconomic con-

EXHIBIT 4

Implied Asset Value (\$ million) and Default Boundary of Seitel Inc.



ditions do not drive out the effect of LT recovery on the market recovery rate. Moreover, the addition of control variables increases the overall explanatory power of the regression with LT recovery by only 4%.

Prepackaged bankruptcies are different from traditional Chapter 11 cases in terms of the filing decision. The debtor and creditors work out a plan of reorganization before filing for Chapter 11. The decision for filing bankruptcy may be made long before the actual filing date. Further, prepackaged cases are less complex, usually associated with a shorter length of restructuring time and thus lower bankruptcy costs. Therefore, the LT model with usual assumptions may not apply to prepackaged cases. In Columns (4) to (6) of Exhibit 5, we repeat our regression analyses after deleting prepackaged filings. We find that both the coefficient on LT recovery and the adjusted R-squared are slightly higher.

Among other control variables, we find that the coefficient of ROA is positive and statistically significant in all specifications, in line with the fact that credi-

tors are likely to recover more in more profitable firms. Negative industry distress dummy values and the positive coefficient on S&P 500 index returns indicate that both the current industry distress environment and macroeconomic prospect play important roles in firm recovery, consistent with the conclusions in Acharya et al. [2007]. The coefficient of fraud is negative and statistically significant across all regressions.

In unreported panels, we control for stock returns during the one year period before Chapter 11 filing and the stock prices at five days before filing. If the explanatory power of the LT model on debt recovery is purely due to the information contained in equity price, then equity returns or prices should provide equal explanatory power in debt recovery. That being said, we do not find such evidence. Further, although the firm capital structure is assumed to be time-independent in the LT model, it may change over time. We therefore define an alternative measure of recovery as the average of LT recovery over the $[-252, -5]$ trading days prior to Chapter 11 filing

EXHIBIT 5

Regression Analysis on Market Recovery

Variables in this exhibit are described in Exhibit 3. Industry fixed effects are controlled for in all regressions. Robust standard errors are reported in square brackets.

	Whole Sample			Excluding Prepackaged Cases		
	(1)	(2)	(3)	(4)	(5)	(6)
LT recovery		0.505*** [0.092]	0.440*** [0.106]		0.571*** [0.121]	0.529*** [0.139]
Number of issues	−0.001 [0.004]		−0.002 [0.004]	0.002 [0.004]		0.001 [0.004]
Log (asset)	0.022 [0.017]		0.027 [0.016]	0.011 [0.020]		0.017 [0.019]
ROA	0.040*** [0.013]		0.027** [0.013]	0.035*** [0.013]		0.021 [0.013]
Leverage	−0.007 [0.035]		−0.018 [0.035]	−0.054 [0.040]		−0.066* [0.039]
Tangibility	−0.106 [0.065]		−0.121* [0.062]	−0.073 [0.073]		−0.099 [0.071]
Industry distress	−0.076* [0.041]		−0.050 [0.040]	−0.083* [0.047]		−0.048 [0.045]
SP500	0.137* [0.070]		0.134* [0.068]	0.143 [0.088]		0.156* [0.086]
Fraud	−0.190*** [0.047]		−0.146*** [0.049]	−0.193*** [0.058]		−0.124** [0.057]
Constant	0.361*** [0.121]	0.205*** [0.056]	0.115 [0.126]	0.450*** [0.149]	0.187** [0.078]	0.156 [0.160]
# of observations	276	276	276	195	195	195
Adjusted R ²	0.181	0.206	0.246	0.181	0.227	0.256

Note: ***, **, * represent statistical significance at the 1%, 5%, and 10% level, respectively.

for each firm. The mean and median of this average expected recovery are very close to those based on the last observation prior to filing, suggesting that the majority of these firms keep a rather stable capital structure over the four quarters before Chapter 11 filing.

We then use the average of LT recovery in all regressions and find similar results. Furthermore, we re-estimate the expected recovery from the LT model with the deadweight costs at 10% and 20%, respectively, and find that the coefficient of LT recovery on the firm-level recovery presented in Exhibit 5 hardly changes. Our results strongly suggest that the recovery derived from the LT model serves as an informative and useful tool in predicting firm recovery post-bankruptcy.

Additional Tests

If the strong predictive power of the LT model is due to its identification within the manager's role of determining the bankruptcy barrier, then factors that create distortions to the endogenously determined

default boundary are expected to weaken the explanatory power of the model. In this section, we consider two situations for which the assumptions of the model are violated, namely, when there are agency problems and heterogeneous bankruptcy costs.

Agency Problems

In their seminal work, Jensen and Meckling [1976] outline that the interests of managers may not be aligned with that of shareholders, thus creating agency problems. Conversely, Leland and Toft [1996] do not consider such principal-agent conflicts in their model. That being said, the existence of agency problems may invalidate the default barrier chosen by managers. The LT recovery, therefore, should have less explanatory power on market recovery when agency problem is more severe.

According to Murphy [1999], "Stock ownership provides the most direct link between shareholder and CEO wealth." The interests of managers are better aligned with the interests of shareholders when managers

hold a large proportion of equity and options, as their wealth is closely tied to the stock performance. We propose two measures on the link between the interests of managers and those of shareholders, namely, CEO stockholdings and the wealth change sensitivities from stock- and optionholdings based on Core and Guay [1999].

We manually collect data on CEO stockholdings, newly granted stocks options, and the amount and value of unexercised exercisable and unexercisable stock options for the fiscal year prior to Chapter 11 filing from the Execucomp database, proxy statements, and 10-Ks through EDGAR. The delta of unexercised options is estimated based on the Black–Scholes formula and the Core and Guay approach. Delta is used to measure the wealth change sensitivities of CEO option- and stockholdings. The expected stock return volatility is measured as the annualized standard deviation of daily stock returns over the 120 trading days preceding the end of the fiscal year. Treasury bond yields are used as the risk-

free interest rate. The CEO wealth delta is the sum of deltas for the exercisable and unexercisable options and the delta of the shareholdings, which are defined as the product of shares owned and the price at the end of fiscal year times 0.01.¹⁸ Delta thus measures the change of CEO wealth in equityholdings per 1% change of stock price.

We are able to collect CEO stockholdings for 222 firms of our final sample. The mean percentage stockholding is 5.2%, higher than the mean of all firms in the Execucomp database for the same time period. The wealth change sensitivities (delta) are estimated for 216 of our sample firms. The mean of the option and stock delta is 193, suggesting that CEOs' wealth changes by an average of \$193,000 per a 1% change of stock price. In other words, CEOs' wealth is closely connected with their stock performance, on average. However, the mean of delta of our sample firms is lower than that of all firms in the Execucomp database for the time period covered in this study.

We separate our sample firms into subgroups by using the median of CEO stockholding and the median of CEO wealth change sensitivity, namely, above-median ownership firms, below-median ownership firms, above-median delta firms, and below-median delta firms. The link between managers' interests and shareholders' interests should be stronger for above-median ownership firms and above-median delta firms. The LT recovery should have stronger explanatory power on market recovery for these two subsamples of firms.

Exhibit 6 presents the results. The coefficient of LT recovery is much lower in Columns (2) and (4) than Columns (1) and (3), respectively. We find that the LT recovery coefficient is as large as 0.74 in Column (3), where the subsample is composed of firms with an above-median delta. This suggests that the market recovery is closer to the recovery derived from the optimal default barrier determined by managers when agency problems are less severe. On the other hand, the expected recovery has less explanatory power on market recovery when CEOs hold less equity and when their

EXHIBIT 6

Agency Conflict and Market Recovery

Variables in this exhibit are described in Exhibit 3. Subsample firms are grouped based on CEO stock ownership and CEO wealth-change sensitivities (delta). Industry fixed effects are controlled for in all regressions. Robust standard errors are reported in square brackets.

	(1) Above median ownership	(2) Below median ownership	(3) Above median delta	(4) Below median delta
LT recovery	0.592*** [0.210]	0.464*** [0.136]	0.737*** [0.176]	0.280 [0.173]
Number of issues	-0.003 [0.009]	-0.002 [0.005]	-0.003 [0.005]	-0.017* [0.009]
Log (asset)	0.023 [0.035]	0.023 [0.021]	0.018 [0.029]	0.062** [0.027]
ROA	0.049*** [0.016]	0.016 [0.024]	0.053*** [0.016]	0.018 [0.024]
Leverage	-0.102 [0.100]	0.011 [0.043]	-0.113 [0.097]	0.069 [0.048]
Tangibility	-0.198 [0.123]	-0.050 [0.092]	-0.123 [0.090]	-0.118 [0.115]
Industry distress	-0.087 [0.053]	0.040 [0.073]	-0.104* [0.056]	0.024 [0.071]
SP500	0.297*** [0.112]	0.092 [0.095]	0.250* [0.126]	0.106 [0.099]
Fraud	-0.137* [0.072]	0.016 [0.089]	-0.101 [0.072]	-0.028 [0.119]
Constant	0.182 [0.241]	0.043 [0.166]	0.047 [0.289]	-0.047 [0.186]
# of observations	111	111	108	108
Adjusted R ²	0.236	0.291	0.345	0.179

Note: ***, **, * represent statistical significance at the 1%, 5%, and 10% level, respectively.

wealth is less tied to the performance of equity. Our finding suggests that in the absence of agency problems, the LT model can serve as a reasonable benchmark for estimating this expected recovery. The presence of agency problems weakens such explanatory power.

Bankruptcy costs. The LT model assumes that bankruptcy costs are fixed for all firms. The constant deadweight costs represent the ex ante bankruptcy costs assumed by managers in determining the default barrier. However, the realized bankruptcy costs incurred in Chapter 11 can be substantially heterogeneous among firms.

For example, Duffie and Lando [2001] suggest that around the time of bankruptcy, substantial information about issuers is revealed to the market. Firm asset value can make a large dive to drop below default threshold because of a sudden increase in expected bankruptcy costs. Meanwhile, the firm asset value may also jump above the default barrier determined by shareholders after the bankruptcy filing, possibly due to hedge fund involvement on the debt side (Jiang et al. [2012]). Therefore, within a short time period after Chapter 11 filing, investors can change their views on the value of the firm and its liabilities dramatically, and the bond price can vary accordingly. As a result, the trading prices measured at the end of the month of bankruptcy filing could differ significantly from the original expected recovery.

In this section, we examine whether the large difference between the realized bankruptcy costs and the ex ante parameter assigned to the model weakens the explanatory power of the model. LoPucki and Doherty [2008] show that legal costs account for a sizeable proportion of the total costs of bankruptcy. Lengthy negotiations among all parties are associated with higher legal fees paid. Indirect costs of bankruptcy, such as the loss of sales, can also be large. Therefore, an anticipation of a likely delay in bankruptcy resolution can weaken the bargaining power of creditors against equityholders as well as management and can thus result in additional distress costs. The duration of bankruptcy, measured as the number of years from the date of filing to the date of plan confirmation or conversion to Chapter 7, is used as a proxy for realized bankruptcy costs. Since a 15% bankruptcy cost is the midpoint of the bankruptcy loss range stated in prior literature, the LT recovery based on this amount for all firms should explain the market recovery better when the realized bankruptcy costs are moderate.

We separate our firms into two groups by using bankruptcy duration as the dividing factor. The reso-

lution length of the first group lies between the 25th and 75th percentile of the duration spectrum—that is, between six months and two years. The second group has a duration of either less than six months or more than two years. We refer to the duration of the first group as moderate and to that of the second group as extreme. Results in Exhibit 7 show that the coefficient of LT recovery is 0.61 for the moderate duration group, much higher than the 0.38 for the extreme duration group. This finding confirms our conjecture that with a moderate ex ante bankruptcy cost parameter assigned to the model, the explanatory power of LT recovery is stronger when the realized bankruptcy costs are in the expected range.

In addition, we adopt a regression model to endogenize the bankruptcy cost parameter assigned to each firm. The dependent variable is the duration of reorganization and the control variables include the logarithm

EXHIBIT 7

Heterogeneous Bankruptcy Costs and Market Recovery

Variables in this exhibit are described in Exhibit 3. Subsample firms are grouped based on the length of time spent in Chapter 11. Industry fixed effects are controlled for in all regressions. Robust standard errors are reported in square brackets.

	(1) Cases with moderate duration	(2) Cases with extreme duration
LT recovery	0.611*** [0.173]	0.377*** [0.141]
Number of issues	0.002 [0.004]	-0.012** [0.006]
Log (asset)	0.016 [0.026]	0.041** [0.020]
ROA	0.031** [0.013]	0.008 [0.031]
Leverage	-0.093 [0.058]	0.077* [0.043]
Tangibility	-0.109 [0.087]	-0.083 [0.094]
Industry distress	-0.063 [0.048]	-0.043 [0.063]
SP500	0.133 [0.100]	0.096 [0.105]
Fraud	-0.214*** [0.062]	-0.039 [0.077]
Constant	0.137 [0.197]	0.027 [0.165]
# of observations	138	138
Adjusted R ²	0.289	0.226

Note: ***, **, * represent statistical significance at the 1%, 5%, and 10% level, respectively.

of the number of debt issues, firm size, profitability, and tangibility. Based on the predicted duration from the regression model, firms are separated into three groups. We re-estimate the LT recovery by assigning a 10% loss to the group of firms with duration in the bottom one-third, 15% to the group with duration in the middle one-third, and 20% to the group with duration in the top one-third. Our multivariate results with the modified LT recovery based on the endogenized bankruptcy costs remain qualitatively the same.

Overall, our results suggest that the endogenous bankruptcy model developed by Leland and Toft [1996] has strong explanatory power in debt recovery. The explanatory power is higher for firms with less severe agency conflict and with moderate bankruptcy costs. Our evidence suggests that the explanatory power of the LT model may derive from its recognition of the manager's role in endogenizing the default barrier. The limitation of this study, however, is that it considers only one specific endogenous bankruptcy model. Other endogenous or strategic default models may possibly produce similar results. Our article calls for future studies on incorporating other structural models to study debt recovery.

CONCLUSION

In this study, we use the Leland–Toft endogenous bankruptcy model to estimate firm recovery through MLE for a large sample of Chapter 11 filings. In their model, bankruptcy is triggered when the firm asset value reaches an endogenously determined barrier level. This model allows us to infer the expected recovery from equity prices and capital structure information. We show that the expected recovery estimated from this model has strong explanatory power on debt recovery observed at the month-end after Chapter 11 filing for our sample firms. We demonstrate that the estimated recovery contains as much information as other sets of control variables used in previous research on debt recovery. Furthermore, our empirical results indicate that agency problems, which are not considered in the Leland–Toft model, might influence the decision for bankruptcy filing and weaken the explanatory power of the estimated recovery. The other factor that weakens the model's explanatory power is the difference between ex ante and realized bankruptcy costs. The results presented in this article provide risk managers and portfolio managers with a useful tool for estimating firm recovery upon bankruptcy filing.

APPENDIX A

Leland–Toft [1996] Model

In the Leland–Toft model, the firm asset value is unobservable and is assumed to follow a geometric Brownian motion:

$$\frac{dV}{V} = (\mu_v - \delta)dt + \sigma_v dW$$

where μ_v is the asset mean rate of return, δ is the constant dividend payout rate, σ_v is the volatility, and W is a Brownian motion. The firm continuously issues a constant principal amount of new debt with a maturity of T years and redeems it at par upon maturity. Debt pays constant coupons each year. The firm's operating cash flow may at times fall short of required current debt payments. As long as the equity option value is higher than the required debt payment, equityholders will sell newly issued stocks to cover the cash shortage and continue to meet debt obligations. However, when the asset values fall below a threshold and new funds cannot be raised, bankruptcy is declared. At bankruptcy, the remaining assets are liquidated and distributed to debtholders after the deduction of bankruptcy costs.

The bond value is expressed as

$$B(V; V^*, T) = F \left[\frac{c}{r} + \left(1 - \frac{c}{r} \right) \left(\frac{1 - e^{-rT}}{rT} - I(T) \right) + \left((1 - w) \frac{V^*}{F} - \frac{c}{r} \right) J(T) \right]$$

where

$$\begin{aligned}
I(T) &= \frac{1}{rT} [G(T) - e^{-rT} \tilde{F}(T)] \\
J(T) &= \frac{1}{z\sigma_v \sqrt{T}} \left[-\left(\frac{V}{V^*}\right)^{-a+z} N(q_1(T)) q_1(T) + \left(\frac{V}{V^*}\right)^{-a-z} N(q_2(T)) q_2(T) \right] \\
G(T) &= \left(\frac{V}{V^*}\right)^{-a+z} N(q_1(T)) + \left(\frac{V}{V^*}\right)^{-a-z} N(q_2(T)) \\
\tilde{F}(T) &= N(h_1(T)) + \left(\frac{V}{V^*}\right)^{-2a} N(h_2(T)) \\
a &= \frac{r-\delta}{\sigma_v^2} - \frac{1}{2}, \quad b = \ln\left(\frac{V}{V^*}\right), \quad z = \sqrt{a^2 + \frac{2r}{\sigma_v^2}} \\
q_1(T) &= \frac{-b - z\sigma_v^2 T}{\sigma_v \sqrt{T}}, \quad q_2(T) = \frac{-b + z\sigma_v^2 T}{\sigma_v \sqrt{T}} \\
h_1(T) &= \frac{-b - a\sigma_v^2 T}{\sigma_v \sqrt{T}}, \quad h_2(T) = \frac{-b + a\sigma_v^2 T}{\sigma_v \sqrt{T}} \\
V^* &= \frac{(c/r)(A/rT - B) - A/rT - \tau c(a+z)/r}{1 + w(a+z) - (1-w)B} F \\
A &= 2ae^{-rT} N(a\sigma_v \sqrt{T}) - 2zN(z\sigma_v \sqrt{T}) - \frac{2}{\sigma_v \sqrt{T}} n(z\sigma_v \sqrt{T}) + \frac{2e^{-rT}}{\sigma_v \sqrt{T}} n(a\sigma_v \sqrt{T}) + z - a \\
B &= -\left(2z + \frac{2}{z\sigma_v^2 T}\right) N(z\sigma_v \sqrt{T}) - \frac{2}{\sigma_v \sqrt{T}} n(z\sigma_v \sqrt{T}) + z - a + \frac{1}{z\sigma_v^2 T}
\end{aligned}$$

$N(\cdot)$ is the cumulative distribution function of the standard normal distribution, and $n(\cdot)$ represents the standard normal density function.

The total market value of the firm is the unlevered asset value plus the value of tax benefits, less the value of bankruptcy costs

$$v(V; V^*, T) = V + \frac{\tau c F}{r} \left[1 - \left(\frac{V}{V^*}\right)^{-(a+z)} \right] - w V^* \left(\frac{V}{V^*}\right)^{-(a+z)}$$

The equity value E is the residual value

$$E(V; V^*, T) = v(V; V^*, T) - B(V; V^*, T)$$

APPENDIX B

Log-Likelihood Function of the Leland–Toft Model

Let $G(\ln V)$ denote the equity pricing function from the structural model. Since the firm unobservable asset value, V , follows a geometric Brownian motion, $\ln V$ follows a normal distribution. The conditional density function of equity can be expressed as

$$\begin{aligned}
\frac{d\text{Prob}(E_t \leq e \mid E_{t-1})}{de} &= \frac{d\text{Prob}[G(\ln V_t) \leq e \mid \ln V_{t-1}]}{de} \\
&= \frac{d\text{Prob}[(\ln V_t) \leq G^{-1}(e) \mid \ln V_{t-1}]}{de} \\
&= N[G^{-1}(e) \mid \ln V_{t-1}] \cdot \frac{dG^{-1}(e)}{de} \\
&= N(\ln V_t \mid \ln V_{t-1}) \cdot \left[\frac{dG(\ln V_t)}{d \ln V_t} \right]^{-1}
\end{aligned}$$

The equation shows that the unknown density function of the equity price can be transformed into a product of a Jacobian function and the density function of firm asset. The log-likelihood function, therefore, can be expressed as

$$\begin{aligned}
L(E_t; t = 1, \dots, N, \mu_v, \sigma_v) &= \ln \left(\prod_{t=2}^N N(\ln V_t \mid \ln V_{t-1}) \cdot \prod_{t=2}^N \left[\frac{dG(\ln V_t)}{d \ln V_t} \right]^{-1} \right) \\
&= \sum_{t=2}^N \ln N(\ln V_t \mid \ln V_{t-1}) - \sum_{t=2}^N \ln \frac{dG(\ln V_t)}{d \ln V_t} \\
&= -\frac{N-1}{2} \ln(2\pi\sigma_v^2) - \frac{1}{2} \sum_{t=2}^N \left[\frac{\ln \frac{V_t}{V_{t-1}} - \left(\mu_v - \frac{\sigma_v^2}{2} \right)}{\sigma_v^2} \right]^2 \\
&\quad - \sum_{t=2}^N \ln \frac{dG(\ln V_t)}{d \ln V_t} \Big|_{\ln V_t = G^{-1}(E_t^{\text{obs}}; \mu_v, \sigma_v)}
\end{aligned}$$

Now, if we replace G with the specific formula in the Leland–Toft model, then the last term of the above equation can be expressed as

$$\begin{aligned}
\frac{dG(\ln V_t)}{d \ln V_t} \Big|_{\ln V_t = G^{-1}(E_t^{\text{obs}}; \mu_v, \sigma_v)} &= \frac{dE_t(V; V^*, T)}{d \ln V_t} \\
&= V_t - \frac{dB(V; V^*, T)}{d \ln V_t}
\end{aligned}$$

where

$$\begin{aligned}
\frac{dB(V; V^*, T)}{d \ln V_t} &= \frac{(r-c)F}{r^2 T} \left[(a-z)N(q_1(T)) + \frac{n(q_1(T))}{\sigma_v \sqrt{T}} \right] \left(\frac{V}{V^*} \right)^{-a+z} \\
&\quad + \frac{(r-c)F}{r^2 T} \left[\frac{n(q_2(T))}{\sigma_v \sqrt{T}} + (a+z)N(q_2(T)) \right] \left(\frac{V}{V^*} \right)^{-a-z} \\
&\quad - \frac{(r-c)F}{r^2 T} \left[2aN(h_2(T)) \left(\frac{V}{V^*} \right)^{-2a} + \frac{n(h_2(T))}{\sigma_v \sqrt{T}} \left(\frac{V}{V^*} \right)^{-2a} + \frac{n(h_1(T))}{\sigma_v \sqrt{T}} \right] e^{-rT} \\
&\quad + \frac{(1-w)V^* - \frac{cF}{r}}{z\sigma_v \sqrt{T}} \left[(a-z)q_1(T)N(q_1(T)) + \frac{q_1(T)n(q_1(T)) + N(q_1(T))}{\sigma_v \sqrt{T}} \right] \left(\frac{V}{V^*} \right)^{-a+z} \\
&\quad - \frac{(1-w)V^* - \frac{cF}{r}}{z\sigma_v \sqrt{T}} \left[(a+z)q_2(T)N(q_2(T)) + \frac{q_2(T)n(q_2(T)) + N(q_2(T))}{\sigma_v \sqrt{T}} \right] \left(\frac{V}{V^*} \right)^{-a-z}
\end{aligned}$$

ENDNOTES

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¹A partial list of this literature includes Altman and Kishore [1996] and Wang [2011] documenting debt seniority and industry type as the most critical determinants of debt recovery; Franks and Torous [1994] finding that the recovery rate is based largely on the book value of securities received in distressed exchange or bankruptcy; Bris et al. [2006] identifying the effect of bankruptcy procedure (Chapter 11 vs. Chapter 7) on debt recovery; Acharya et al. [2007] showing industry distress at the time of default as crucial determinant; and Carey and Gordy [2009] suggesting the effect of the proportion of bank loans on firm-level recovery.

²For example, Pan and Singleton [2008] use credit default spread on a panel of sovereign countries to jointly identify recovery rates and default intensity, while Das and Hanouna [2009] and Elkamhi et al. [2010] extract default probability and recovery rates using the term structure of corporate credit default swap spreads. Unal et al. [2003] use corporate debt to estimate implied recovery rate.

³Eom et al. [2004] suggest that endogenous default barrier models outperform other models at generating the smallest bond pricing errors.

⁴Although Chapter 11 can, at times, be filed by creditors against the debtor, section 303(h) (1) of the U.S. Bankruptcy Code makes it difficult to force a firm into bankruptcy. To force a company into bankruptcy, creditors must show a "general pattern of nonpayment of undisputed mature debts during the period immediately preceding the petition." A wide range of factors are considered for this requirement. These factors include the percentage of debt in default, the length of time the debt is overdue, the increasing size of overdue debt over time, and so on. Moreover, three or more creditors must join the involuntary bankruptcy petition for a firm with 12 or more creditors. With respect to this rule, the courts also carefully scrutinize cases whose creditors are closely related parties, or whose claim has been sold to parties who later join the bankruptcy petition.

⁵Strategic debt service models also consider shareholders' decision in determining bankruptcy filing. However, we do not consider such models in this study for two reasons. First, a disadvantaged feature of these models is that the closed-form solution is derived if the firm has perpetual debt and the analytical solution is not available for a firm with finite debt maturity (see Fan and Sundaresan [2000]). Second, empirical evidence (thus far) is not in favor of strategic default models in explaining bond yield spreads (see Huang and Huang [2012]).

⁶If the value of collateral is less than the face value of secured bank debt, the proportion that exceeds the collateral joins the unsecured debt to share the remaining asset. Davydenko and Franks [2008] show that collateral has strong predictive power on bank debt recovery.

⁷One of the possible limitations of the MLE method is that it estimates a constant drift term and a constant asset volatility for the given estimation period. However, existing structural models do not specify the drift term to be constant. Furthermore, constant volatility assumption is shown to play a potential role in structural models' inability to capture the dynamic behavioral of credit spreads (see Huang and Zhou [2008]).

⁸The LT model assumes that the firm continuously sells a constant amount of new debt with initial maturity of T years from issuance, which (if solvent) it will redeem at par upon maturity. Therefore, at the beginning of each issuance, the time to maturity of firm debt is uniformly distributed in the range of $[0, T]$ and the weighted-average maturity of firm debt is $T/2$. Since bank loan maturity information is largely unavailable, we adopt one year, following Davydenko [2012], who shows that using the weighted-average maturity of corporate bonds rather than one year for bank loans does not influence the LT estimation results.

⁹Standard and Poor's [2009] treats administrative expenses, which are related to professional fees and other costs associated with bankruptcy, as 3%–10% of the borrower's enterprise value, depending on the complexity of the capital structure. This percentage represents the direct costs. With a similar amount of indirect costs, our assumption of 15% bankruptcy cost is reasonable. As a robustness check, we also estimate the LT model with 10% and 20% deadweight loss and find our results hold qualitatively the same.

¹⁰Detailed information can be found at www.federal-reserve.gov.

¹¹Financial institutions keep high leverage. Their bankruptcy filing decisions may not be determined through an endogenous default barrier.

¹²The LossStats database was developed by S&P to capture defaults and debt recoveries of firms with more than \$50 million of debt outstanding at default from 1987. It records the

average trading price of defaulted corporate debt, including bank loans, from 15 to 45 days after default to measure debt recovery upon default.

¹³New Generation Research collects distressed security trading prices from various sources, including court documents, company filings, and Bloomberg.

¹⁴Post-default distress trading price based on par value is treated as a standard measure of debt recovery at default. A number of prior empirical studies (see Altman and Kishore [1996] and Cantor et al. [2009]) use the 30-day post-default bid prices as a percentage of face value as a measure of recovery. Eberhart and Sweeney [1992] document that bond prices at the time of bankruptcy are unbiased estimates of the bonds' payoff upon settlement of the bankruptcy.

¹⁵See Guha [2003] and Cantor et al. [2009].

¹⁶Ericsson and Reneby [2005] demonstrate that the MLE method is able to provide unbiased structural model parameters for the 90-day equity value sample, although increasing the equity sample size leads to improved efficiency and normality.

¹⁷Alternatively, we define the industry distress dummy based on the three-digit SIC and turn to the two-digit SIC if there are fewer than five firms in a three-digit SIC category in a given year. The results are very similar when using this alternative industry classification.

¹⁸Refer to Core and Guay [1999] for details on estimating the delta of option- and stockholdings.

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